Predicting Stunting Percentage in Children Under Five Using Maternal Literacy and Prenatal Care Access

SDG Goal: SDG 2 – Zero Hunger

Cohort Number: 17

Submission Date: 3rd March of 2026

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ABSTRACT

This project analyzes the relationship between maternal literacy, prenatal care access, and child stunting prevalence using data sourced from the World Bank's World Development Indicators (WDI) database. Relevant indicators including stunting prevalence (% of children under five), female literacy rate (%), and prenatal care coverage (%) were extracted and compiled into a structured dataset. The dataset was cleaned and preprocessed prior to analysis. Missing categorical values were handled using mode imputation to ensure data completeness. Exploratory Data Analysis (EDA) was conducted to examine distribution patterns and relationships between variables before proceeding to regression modeling. The study contributes to understanding how maternal education and healthcare access influence child malnutrition, supporting Sustainable Development Goal 2 (Zero Hunger).

1. INTRODUCTION

Child stunting is an indicator of chronic malnutrition and reflects long-term nutritional deprivation during early childhood. It significantly affects physical growth, cognitive development, and long-term productivity. Reducing stunting remains a critical public health objective, especially in developing regions.

Maternal literacy plays a vital role in improving child health outcomes, as educated mothers are more likely to adopt improved nutrition practices, seek healthcare services, and maintain better hygiene standards. Similarly, access to prenatal care ensures early monitoring of pregnancy, nutritional counseling, and preventive healthcare interventions that directly influence child growth outcomes.

This study applies a data-driven analytical approach to examine the impact of maternal literacy and prenatal care coverage on stunting prevalence.

2. PROBLEM STATEMENT

The objective of this project is to predict the stunting percentage in children under five based on maternal literacy and access to prenatal care, while handling missing categorical data using mode imputation.

3. SDG IMPACT ANALYSIS

This project aligns with Sustainable Development Goal 2: Zero Hunger, which aims to end all forms of malnutrition. Stunting reduction is a measurable indicator of progress toward this goal. By analyzing the relationship between maternal literacy, prenatal care access, and stunting prevalence, this study provides insights that may support evidence-based public health policies and targeted interventions aimed at reducing malnutrition and improving long-term socio-economic development.

4. DATASET DESCRIPTION

4.1 Data Source

The dataset was sourced from the World Bank's World Development Indicators (WDI) database, which provides internationally comparable statistics on development, education, and health indicators.

Source:
https://databank.worldbank.org/source/world-development-indicators

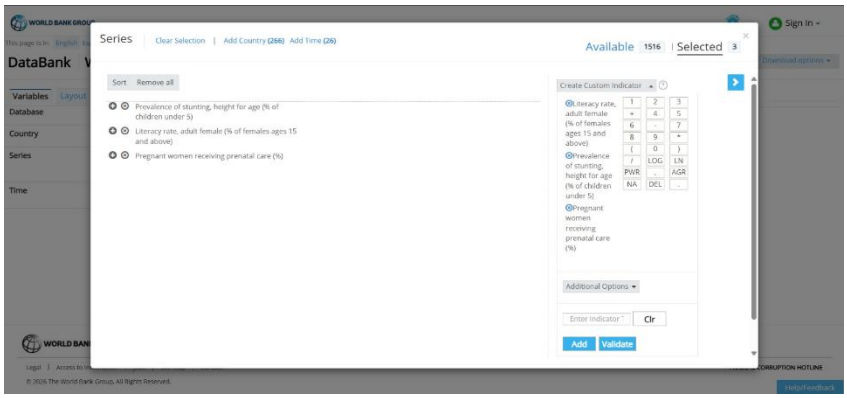


Figure 1(a): Selected Indicators from World Bank (WDI)

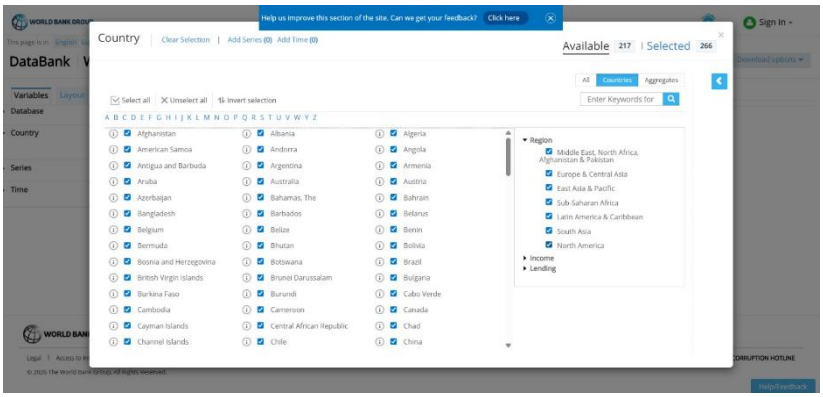


Figure 1(b): Country Selection in WDI Database

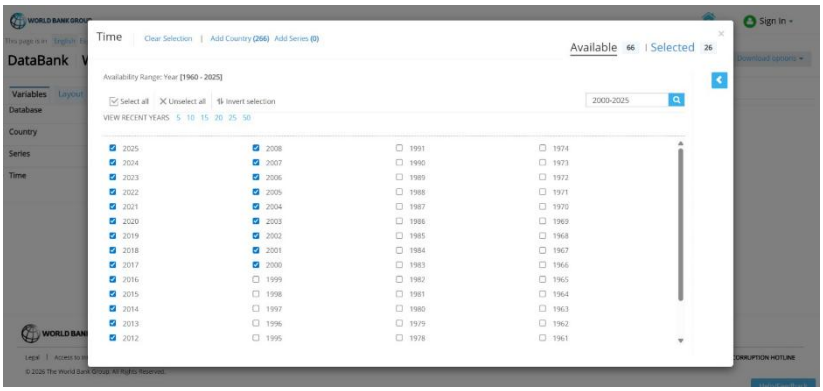


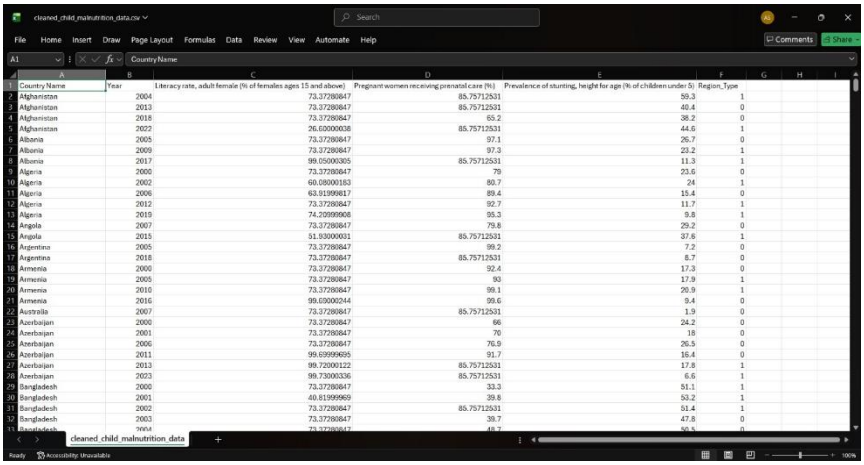
Figure 1(c): Time Period Selection in WDI Database

The selected indicators include:

- Prevalence of stunting, height-for-age (% of children under five)
- Female literacy rate (%)
- Prenatal care coverage (%)

4.2 Raw Dataset (CSV Compilation)

The extracted indicators were compiled into a structured CSV file for analysis.



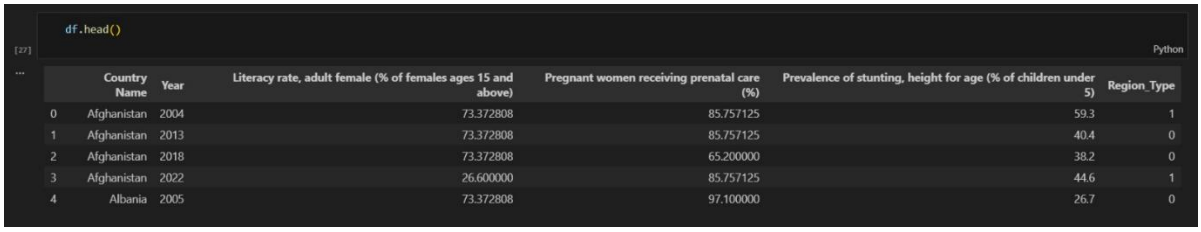
Country Name	Year	Literacy rate, adult female (% of females ages 15 and above)	Pregnant women receiving prenatal care (%)	Prevalence of stunting, height for age (% of children under 5)	Region_Type
Afghanistan	2004	73.37280847	85.75712531	59.3	1
Afghanistan	2013	73.37280847	85.75712531	40.4	0
Afghanistan	2018	73.37280847	65.2	38.2	0
Afghanistan	2022	26.60000038	85.75712531	44.6	1
Albania	2005	73.37280847	97.1	26.7	0
Albania	2009	73.37280847	97.3	23.2	1
Albania	2017	99.05000305	85.75712531	11.3	1
Algeria	2000	73.37280847	79	23.6	0
Algeria	2002	60.08000183	80.7	24	1
Algeria	2006	63.91999817	89.4	15.4	0
Algeria	2012	73.37280847	92.7	11.7	1
Algeria	2019	74.20999908	95.3	9.8	1
Angola	2007	73.37280847	79.8	29.2	0
Angola	2015	51.93000031	85.75712531	37.6	1
Argentina	2005	73.37280847	99.2	7.2	0
Argentina	2018	73.37280847	85.75712531	8.7	0
Armenia	2000	73.37280847	92.4	17.3	0
Armenia	2005	73.37280847	93	17.9	1
Armenia	2010	73.37280847	99.1	20.9	1
Armenia	2016	99.69000244	99.6	9.4	0
Australia	2007	73.37280847	85.75712531	1.9	0
Azerbaijan	2000	73.37280847	66	24.2	0
Azerbaijan	2001	73.37280847	70	18	0
Azerbaijan	2006	73.37280847	76.9	26.5	0
Azerbaijan	2011	99.69999695	91.7	16.4	0
Azerbaijan	2013	99.72000122	85.75712531	17.8	1
Azerbaijan	2023	99.73000336	85.75712531	6.6	1
Bangladesh	2000	73.37280847	33.3	51.1	1
Bangladesh	2001	40.81999969	39.8	53.2	1
Bangladesh	2002	73.37280847	85.75712531	51.4	1
Bangladesh	2003	73.37280847	39.7	47.8	0

Figure 2: Raw Dataset Preview (CSV Format)

This CSV file served as the input dataset for preprocessing and analysis.

4.3 Dataset Overview

The dataset consists of structured observations representing the selected indicators. The target variable is stunting percentage, which is continuous in nature and suitable for regression analysis. The independent variables include maternal literacy rate and prenatal care coverage.



```
df.head()
```

[27] Python

	Country Name	Year	Literacy rate, adult female (% of females ages 15 and above)	Pregnant women receiving prenatal care (%)	Prevalence of stunting, height for age (% of children under 5)	Region_Type
0	Afghanistan	2004	73.372808	85.757125	59.3	1
1	Afghanistan	2013	73.372808	85.757125	40.4	0
2	Afghanistan	2018	73.372808	65.200000	38.2	0
3	Afghanistan	2022	26.600000	85.757125	44.6	1
4	Albania	2005	73.372808	97.100000	26.7	0

Figure 3: Dataset Preview Using DataFrame (df.head())

The dataset was verified for structure, completeness, and compatibility with statistical modeling.

5. DATA PREPROCESSING

5.1 Missing Value Analysis

Missing value analysis was performed to identify incomplete entries in the dataset.

```
[37]  df.isna().sum()                                                                    Python

...  Country Name                                                    0
     Year                                                            0
     Literacy rate, adult female (% of females ages 15 and above)    0
     Pregnant women receiving prenatal care (%)                      0
     Prevalence of stunting, height for age (% of children under 5)  0
     Region_Type                                                     0
     dtype: int64
```

Figure 4: Missing Value Summary Before Imputation

The analysis indicated missing values in the prenatal care variable.

5.2 Mode Imputation

Since the prenatal care variable is categorical in nature, mode imputation was applied. Mode imputation replaces missing values with the most frequently occurring category in the respective column, preserving distribution characteristics.

```
[37]  df.isna().sum()                                                                    Python

...  Country Name                                                    0
     Year                                                            0
     Literacy rate, adult female (% of females ages 15 and above)    0
     Pregnant women receiving prenatal care (%)                      0
     Prevalence of stunting, height for age (% of children under 5)  0
     Region_Type                                                     0
     dtype: int64
```

Figure 5: Missing Value Summary After Mode Imputation

After imputation, no missing values remained in the dataset.

5.3 Outlier Detection

Outlier detection was conducted using boxplot visualization to assess the presence of extreme values that could influence regression results.

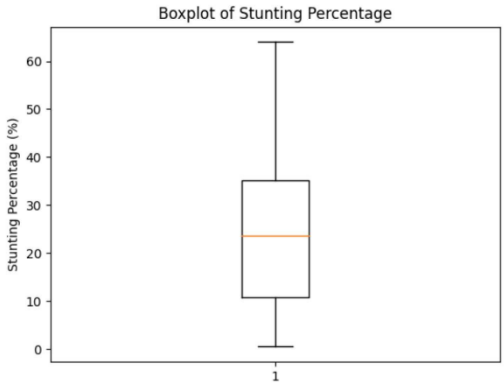


Figure 6: Boxplot for Stunting Percentage

The visualization indicated no extreme outliers significantly affecting dataset stability.

5.4 Encoding and Feature Preparation

Categorical variables were converted into numerical format using appropriate encoding techniques to ensure compatibility with regression modeling. The dataset was then structured by separating independent variables (maternal literacy and prenatal care) from the target variable (stunting percentage).

6. EXPLORATORY DATA ANALYSIS (EDA)

6.1 Distribution of Stunting Percentage

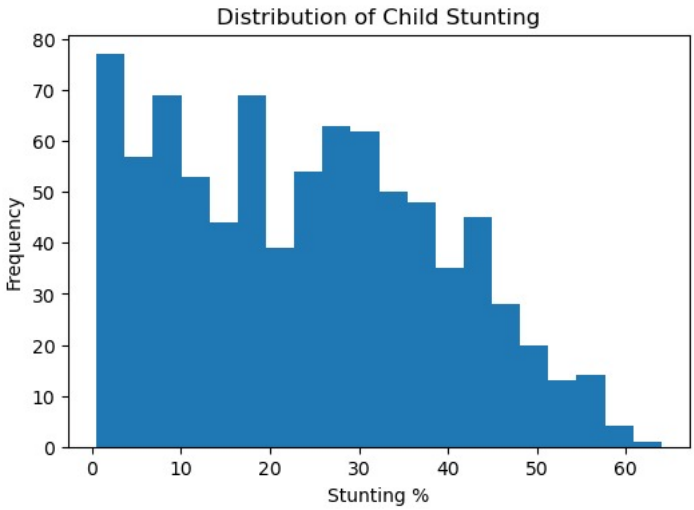


Figure 7: Histogram of Stunting Percentage

The distribution analysis indicates variability in stunting prevalence across observations, supporting the need for predictive modeling.

6.2 Maternal Literacy vs Stunting

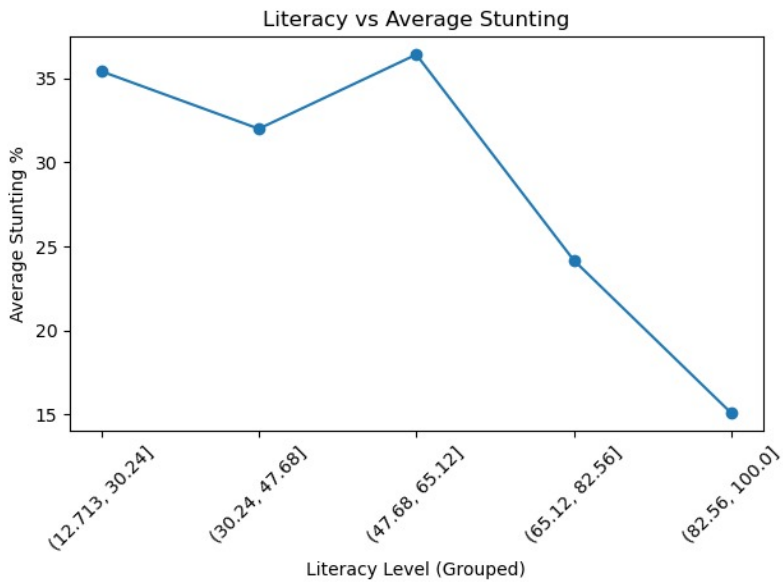


Figure 8: Maternal Literacy vs Stunting Percentage (Scatter Plot)

A negative relationship is observed between maternal literacy and stunting prevalence, suggesting that higher literacy rates correspond to lower stunting percentage

6.3 Prenatal Care vs Stunting

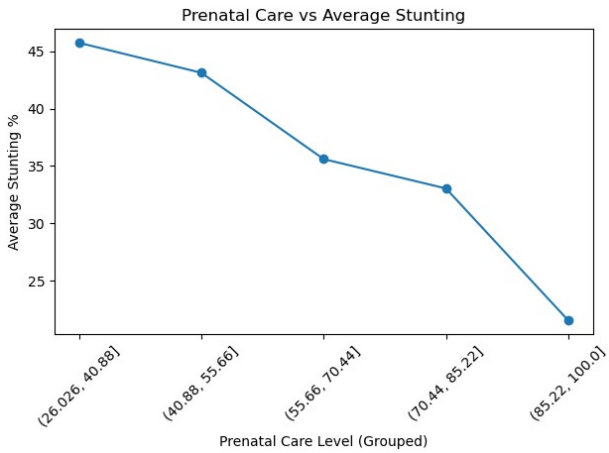


Figure 9: Prenatal Care Coverage vs Stunting Percentage (Scatter Plot)

An inverse relationship is observed between prenatal care coverage and stunting prevalence, indicating that improved access to prenatal care is associated with reduced stunting rates.

6.4 Correlation Analysis

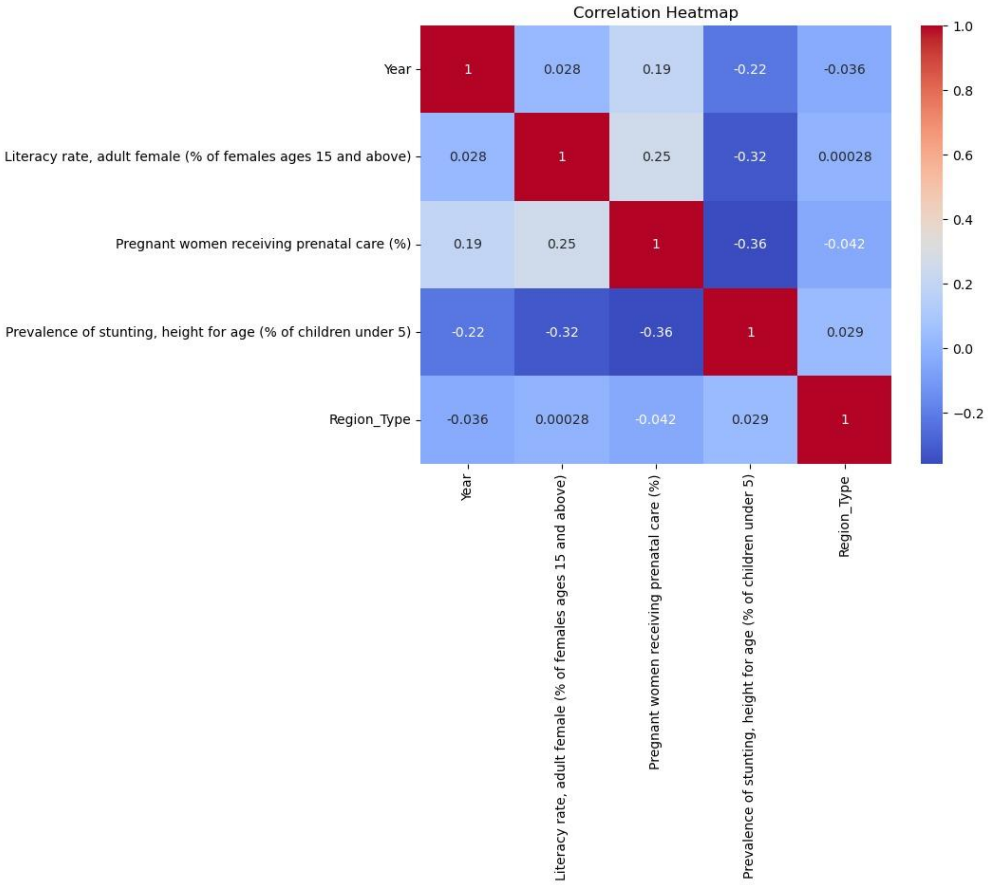


Figure 10: Correlation Matrix Heatmap

The correlation matrix confirms meaningful associations between maternal literacy, prenatal care access, and stunting percentage.

7. MODEL SELECTION & TECHNICAL STRATEGY

Since the target variable, stunting percentage, is continuous in nature, a regression-based modeling approach was selected. Regression models are appropriate for predicting numerical outcomes and analyzing the relationship between independent variables and a continuous dependent variable.

To ensure robust evaluation and avoid reliance on a single approach, multiple regression models were implemented and compared. This allows identification of the most suitable model for the dataset.

The following models were selected:

- Linear Regression
- Decision Tree Regressor
- Random Forest Regressor

Linear Regression was used as a baseline model to evaluate the linear relationship between maternal literacy, prenatal care access, and stunting percentage.

Decision Tree Regressor was included to capture potential non-linear relationships.

Random Forest Regressor, an ensemble learning method, was implemented to improve prediction stability and reduce variance.

8. MODEL IMPLEMENTATION

The dataset was divided into training and testing sets using an 80:20 split ratio. The independent variables included maternal literacy rate and prenatal care coverage, while the dependent variable was stunting percentage.

Each model was trained using the training dataset and evaluated on the testing dataset. The implementation was carried out using the Scikit-learn library in Python.

For each model, the following steps were performed:

- Model initialization
- Model training using the training dataset

- Prediction on the testing dataset
- Performance evaluation using MAE and R^2

This ensured consistent comparison across all models.

9. MODEL EVALUATION

Model performance was evaluated using the following metrics:

- **Mean Absolute Error (MAE):** Measures the average magnitude of prediction errors.
- **R-squared (R^2):** Indicates the proportion of variance in the dependent variable explained by the model.

These metrics were calculated using the test dataset to assess predictive performance.

9.1 Linear Regression Results

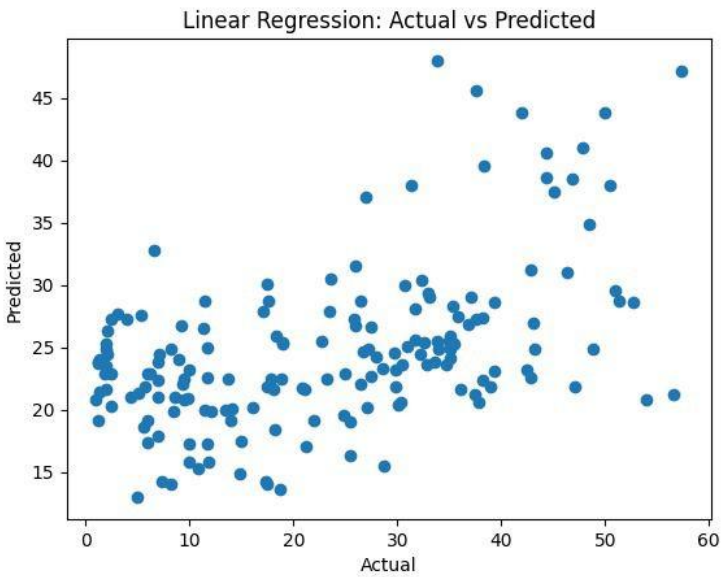


Figure 11: Linear Regression – Actual vs Predicted

The Linear Regression model shows a scattered distribution of predicted values compared to actual values. Predictions appear concentrated within a limited range and tend to underpredict higher stunting percentages.

The visual pattern indicates that the model struggles to accurately capture variability in higher stunting values, suggesting limited ability to model complex relationships between maternal literacy, prenatal care, and stunting percentage.

9.2 Decision Tree Regressor Results

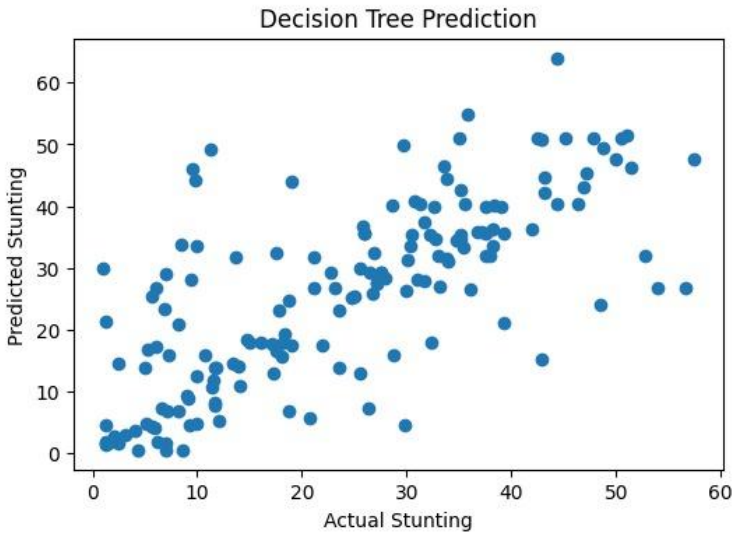


Figure 12: Decision Tree – Actual vs Predicted

The Decision Tree Regressor demonstrates an improved upward trend compared to Linear Regression. Predicted values increase as actual values increase, indicating the model captures non-linear relationships.

However, noticeable dispersion and some extreme predictions are observed, suggesting moderate instability and possible overfitting tendencies.

9.3 Random Forest Regressor Results

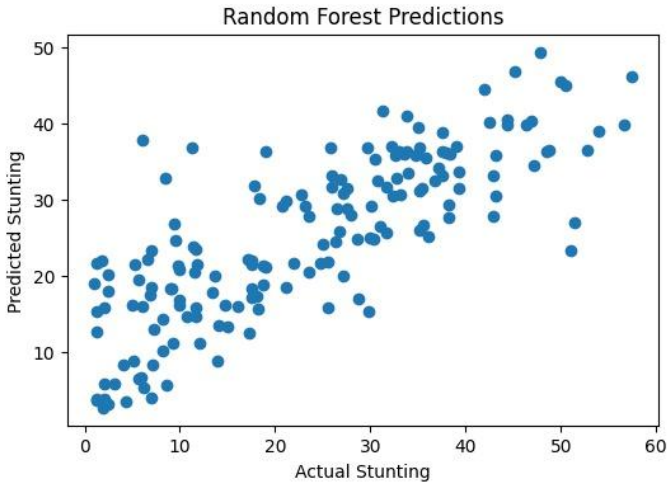


Figure 13: Random Forest – Actual vs Predicted

The Random Forest model exhibits a clearer and more consistent upward trend between actual and predicted values. The points are more closely aligned along the expected direction, indicating stronger predictive performance.

Compared to the other models, Random Forest demonstrates better generalization and reduced variance, making it more reliable for predicting stunting percentage.

10. MODEL COMPARISON

The performance of all three regression models was compared using Mean Absolute Error (MAE) and R-squared (R^2) values obtained from the test dataset.

Model	**MAE**	**R^2**
Linear Regression	10.994333081810574	0.2338700101475073
Decision Tree	7.0727810650887575	0.47462017637887555
Random Forest	6.776491124260356	0.6298109799307027

The results indicate that Random Forest Regressor achieved the lowest MAE and highest R^2 score among the three models.

11. BEST MODEL JUSTIFICATION

Based on the evaluation results, the Random Forest Regressor demonstrated superior predictive performance compared to Linear Regression and Decision Tree models.

With an R^2 score of approximately 0.63, it explains about 63% of the variance in stunting percentage. Additionally, it achieved the lowest MAE (6.78), indicating more accurate predictions.

Therefore, Random Forest Regressor is selected as the final model for predicting stunting percentage.

12. OVERFITTING ANALYSIS

To assess model stability, training and testing performance were compared. Random Forest demonstrated improved performance on the test dataset compared to the baseline models. The

difference between training and testing accuracy was not excessively large, indicating that the model does not exhibit severe overfitting.

This suggests that the Random Forest model generalizes reasonably well to unseen data.

13. RESULTS & KEY INSIGHTS

The modeling results indicate that maternal literacy and prenatal care access significantly influence stunting prevalence.

Linear Regression showed limited performance, suggesting that the relationship between predictors and stunting is not purely linear. Decision Tree improved performance by capturing non-linear patterns, while Random Forest provided the most accurate and stable predictions.

The findings suggest that improvements in maternal education and healthcare access are associated with reduced stunting percentages. This highlights the importance of educational and healthcare interventions in addressing child malnutrition.

14. CONCLUSION

This study successfully developed and evaluated regression models to predict stunting percentage based on maternal literacy and prenatal care access.

Among the models tested, Random Forest Regressor achieved the best performance with the lowest MAE and highest R^2 score. The results indicate that ensemble learning methods provide better predictive capability for this dataset.

The findings support Sustainable Development Goal 2 (Zero Hunger) by demonstrating how maternal education and healthcare access contribute to reducing child malnutrition.

15. FUTURE SCOPE

Future work may include:

- Include additional socio-economic variables
- Expand dataset to more regions and years
- Apply advanced ensemble or boosting models
- Develop an interactive prediction interface

16. TEAM COLLABORATION & MEETING LOGS

Regular group meetings were conducted to ensure systematic progress and collaborative decision-making throughout the project.

Meetings were held on the following dates:

- 26th February 2026
- 27th February 2026
- 28th February 2026
- 1st March 2026

During these meetings, the team discussed dataset selection, preprocessing strategies, exploratory data analysis, model implementation, evaluation metrics, and report structuring.

Meeting – 26th February 2026

Discussion Focus:

- Finalizing dataset selection
- Reviewing variables (maternal literacy, prenatal care, stunting)
- Planning preprocessing steps

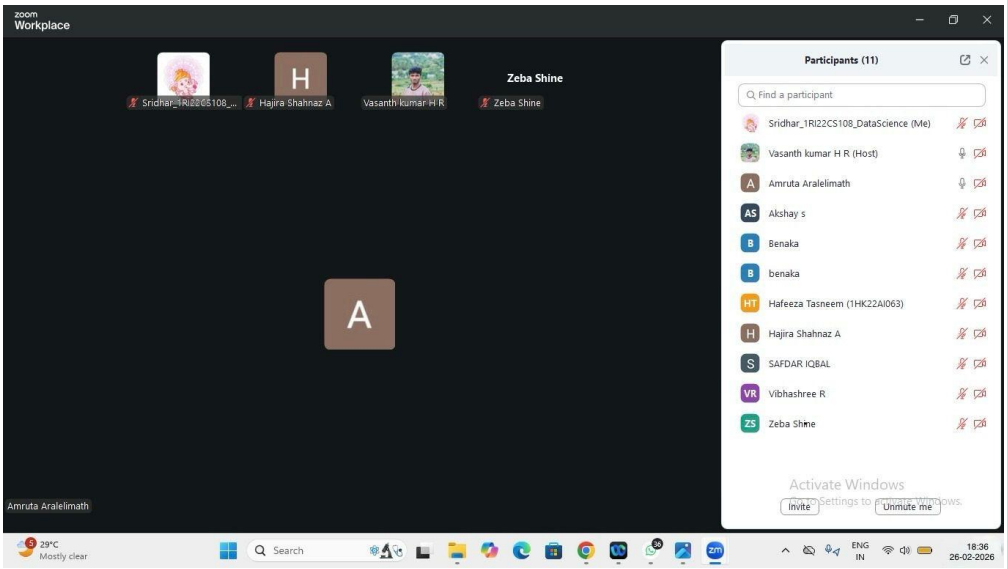


Figure 14: Team Meeting – 26th February 2026

Meeting – 27th February 2026

Discussion Focus:

- Handling missing values using mode imputation
- Reviewing EDA outputs
- Correlation analysis interpretation

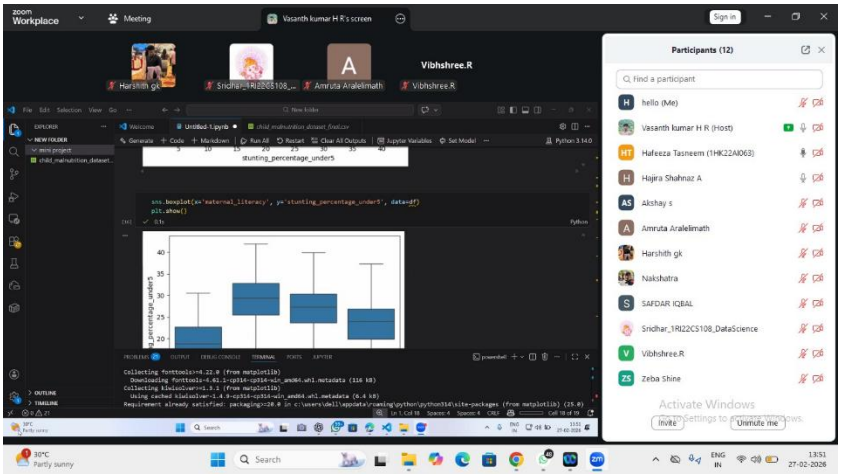


Figure 15: Team Meeting – 27th February 2026

Meeting – 28th

Discussion Focus:

- Training regression models
- Comparing Linear Regression, Decision Tree, and Random Forest
- Reviewing evaluation metrics (MAE and R^2)

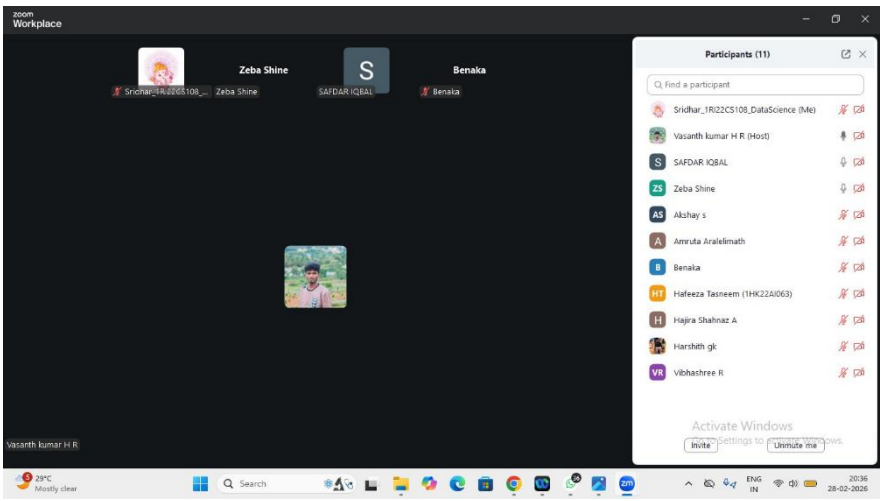


Figure 16: Team Meeting – 28th February 2026

Meeting – 1st March 2026

Discussion Focus:

- **Model comparison**
- **Selecting best-performing model**
- **Finalizing documentation structure**

Figure 17: Team Meeting – 1st March 2026